FIRST: Stale Rollouts Can Pick the Wrong Training Data

A Measurement-First Audit of Off-Policy Data Selection in RLVR

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Abstract

Data selection for reinforcement learning with verifiable rewards (RLVR) studies the problem of choosing which prompts to train on. Because fresh rollouts for every candidate prompt are expensive, existing methods rank prompts with *stale* rollouts generated by an earlier policy, corrected by an importance weight over only one part of the trajectory. The uncorrected part introduces a bias that current practice assumes is negligible, but whether the resulting rankings can actually be trusted has not been examined. This raises two questions: (I) When can a stale ranking be trusted? (II) What should replace it when it cannot? For (I), we prove a no-free-lunch result: either one-sided information set admits statistically indistinguishable pools with opposite true top-k decisions, forcing worst-case selection error at least $1/2$ while standard stale diagnostics remain benign. We then show empirically that whether this failure occurs is decided by one measurable quantity: the reliability of the ranking signal in the prompt pool, which we estimate with a split-half floor. For (II), we propose FIRST (Floor-Informed Reliability of SelecTion), a procedure that reads the artifacts a training run already produces and prescribes, with no new rollouts, one of three policies: uniform sampling, a zero-cost pass-rate ranking, or stale scores with the disagreeing prompts sent to a small audit. On every pool we evaluated, the pre-registered verdict was that no recoverable ranking signal exists, and the zero-cost heuristic matched every gradient-based method. Together, these results suggest that the reliability of the ranking signal should be measured before any selection method is applied or evaluated.

1 Introduction

RLVR pipelines choose their training data by scoring each candidate prompt with "how much would training on it help the current policy π?" and keeping the top k. Estimating target-policy scores requires fresh rollouts from π for every candidate, at every selection round. That is expensive. So practice reuses the rollouts and rewards produced by an earlier behavior policy β, and applies an importance correction. The correction has two axes: (a) how the current policy *reaches* a token (prefix occupancy), and (b) how it *finishes* afterwards (continuation outcome). Several practical estimators correct at most one axis. This is not an oversight: correcting both multiplies hundreds of token ratios, and the variance explodes. The implicit assumption is that the uncorrected half adds only a small bias—small enough to leave the *ranking* intact.

Published evidence already challenges this assumption. The method that popularized single-token influence scoring reports, in its own appendix, two relevant numbers: on- and off-policy per-prompt gradients agree pointwise (cosine above 0.6 on 40 of 50 prompts), yet only 28.8% of

*Draft v0.5 (2026-08-18). Reframed structure following the pre-registered "structural absence" branch; numbers are from completed runs (v1 gate + v2 corrected, 2 seeds); pending seed harvests will update Sections 5 and 8 only.

their top-10% neighborhood structure survives [CROPI]. We treat this as a symptom, not as proof of our mechanism. The number is not decomposed over the two correction axes, and other explanations—validation noise, difficulty clustering, score calibration—can damage rank structure too. This paper supplies what is missing: the mechanism class that produces exactly this dissociation, the decomposition that makes the explanations testable against each other, and a measured map of where the failure does and does not appear.

We audit the assumption end to end. The situation turns out worse, and more structural, than a bias–variance account suggests. Our claims come in three tiers:

1. **Impossibility (proved).** Either one-sided information set is compatible with opposite true top-k decisions, forcing minimax error at least $1/2$ even at arbitrarily small policy divergence (Section 3). Group normalization does not help, and KL, ESS, and stale self-cosine remain benign. Cell disagreement is a useful warning, not a certificate.

2. **Regime map (measured).** Whether the flip actually happens depends on the task–capability regime (Section 5). We chart it with a pre-registered protocol and find three regimes: one where one-sided selection measurably loses precision, one where stale selection tracks the oracle, and one where selection degenerates outright.

3. **Measurement limits (proved + measured).** Exact top-k identification obeys a gap-dependent $\Omega(\sigma^2\Delta^{-2}\log(1/\delta))$ lower bound; our observed boundary gaps place direct certification 5–8 orders of magnitude beyond the measured budget (Section 6). Independently, a decision-theoretic measurement ceiling caps what any oracle-noise-oblivious selector can demonstrate against a noisy oracle (Section 4).

Figure 1 summarizes the framework and how the sections fit together.

The central object of this paper is not an estimator but a measurement: the *split-half oracle floor*. The oracle itself is built from rollouts, so its own ranking carries noise. The floor quantifies that noise: split the oracle's rollouts into two independent halves, rank the prompts with each half, and record the top-k agreement between the two rankings. A low floor means the pool offers little recoverable ranking signal at that rollout budget, and our ceiling result (Proposition 2) shows that no method conditionally independent of the held-out oracle noise can demonstrate more expected precision than the floor permits under the stated model. We applied this standard to our own pools by running FIRST (Floor-Informed Reliability of SelecTion), the pre-registered diagnosis of Section 8, and obtained the verdict *structural absence*: on saturated or tie-degenerate pools, increasing the oracle budget up to K=256 rollouts per prompt is not predicted to create a signal regime. This verdict, rather than a new estimator, is the paper's main empirical finding, and we argue it is the more useful one for practitioners deciding where to spend fresh rollouts.

One further observation applies to every experimental section. Selection evaluation tends to fail in the optimistic direction without producing any visible error. During this project our own pipeline generated four such evaluation artifacts, each of which initially looked like a publishable positive result, and later detected all four (Section 7). We document each mechanism together with its guard, because the same mechanisms can arise in any experiment that reuses rollouts for data selection.

2 Related work

Off-policy influence and data selection. Influence scores built on a single token's importance ratio rank prompts from offline trajectories [CROPI]. In our decomposition this is g_{00}: neither axis

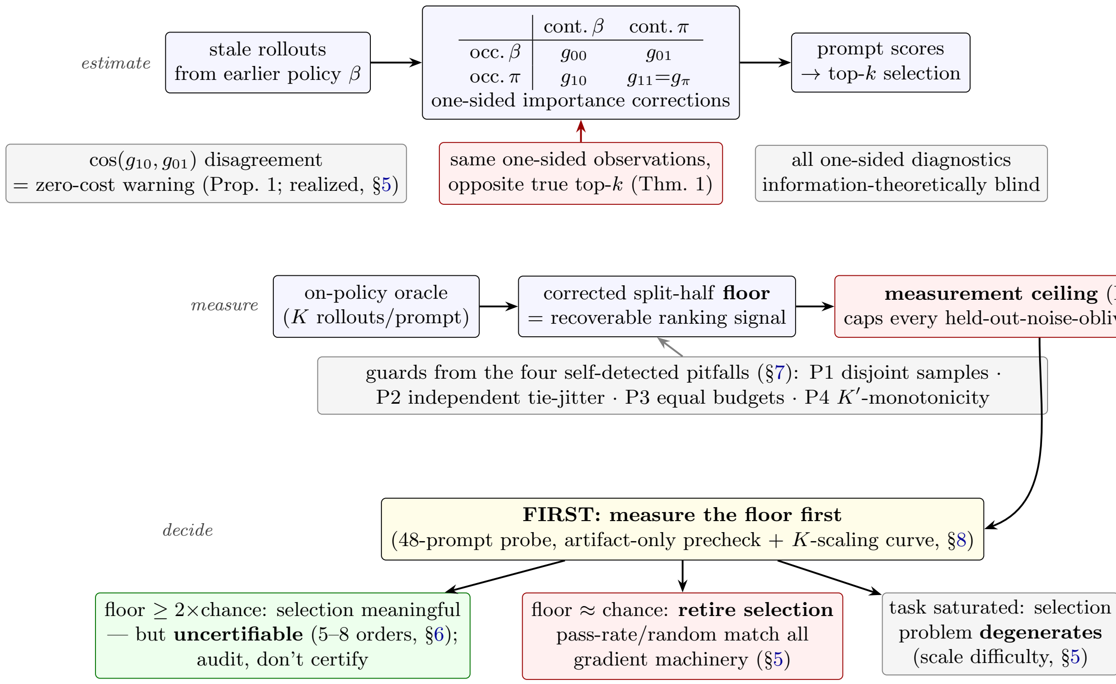

Figure 1: Framework overview. *Estimate:* stale-rollout training-data ranking uses one of four corrections in a 2×2 decomposition; either one-sided information set admits indistinguishable pools with opposite true rankings, so no selector or diagnostic restricted to that information can guarantee safety (§3). *Measure:* the split-half floor of the oracle, computed under four guards, quantifies recoverable signal and induces a ceiling on what any held-out-noise-oblivious method can demonstrate under the parallel-oracle model (§4). *Decide:* an artifact-only, pre-registered diagnosis routes each pool to one of three regimes; in ours the verdict was structural absence (§8).

is restored. The appendix of that work already reports the symptom that motivates our estimand: per-prompt gradients agree pointwise (cosine above 0.6 on 40 of 50 prompts), yet only 28.8% of the top-10% neighborhood structure is preserved. Corrections that reweight the visited prefix [A Step Back, Cumulative] restore the occupancy axis (g_{10}). Their guarantees, however, assume a target-policy advantage, and practical critic-free objectives replace that advantage with a behavior-generated outcome reward; this substitution is the gap our counterexamples use. Corrections that reweight the next $N-1$ sampled tokens [Multi-Step] interpolate toward the outcome-corrected cell (g_{01}); they equal it only when N spans the remaining horizon. Full-trajectory corrections [TIC-GRPO] restore both axes, but their variance grows with response length, so they serve as a cost reference rather than a practical selector. The same tension appeared for Data Shapley in supervised data selection: valuation scores failed to outperform random selection until the selector was redesigned [Rethinking-Shapley, NASH]. Our protocol (§8.1) is the RLVR counterpart, with one difference: the repair comes from a reliability measurement, not from a new estimator.

Rollout allocation and learning stability. Allocation methods [GradAlign, LearnAlign] decide where to spend fresh rollouts to maximize training yield, and stale-data stabilizers keep *learning* well-behaved under reuse. Neither certifies, or even measures, the correctness of the top-k *selection decision*, which is what practitioners actually consume. Classical off-policy correction and subset selection provide the statistical background: conditional importance sampling [Rowland et al.], stationary-distribution correction [Liu et al.], doubly-robust policy gradients [Huang & Jiang], and information-complexity lower bounds for top-k identification [Kaufmann et al.]. Our certification result (Section 6) is an empirical instance of the last of these: when value gaps at the ranking boundary vanish, identification cost diverges. A general budgeted algorithm for top-k certification also exists: it concentrates expensive queries at the ranking boundary and assumes the cheap scores are valid up to noise [ACE]. Our results identify the case this assumption excludes. A stale one-sided score can be systematically biased (sign-flipped), and then certification fails through bias, not through budget. The two failure ingredients we study also appear in adjacent work on on-policy distillation [TIDE]. There, degenerate repetition loops drive token-level agreement to near-zero divergence while outputs collapse, so a pointwise metric misses the object being consumed; and raw token log-ratios are dominated by the extreme tail, with the most negative 1% of tokens carrying roughly half the gradient. Our 0.1% full-correction ESS is an analogous tail-concentration symptom, not the same statistic.

Reliability of noisy rankings. Our floor and ceiling analysis imports split-half reliability and the Spearman–Brown prophecy [Spearman, Brown] into RLVR selection evaluation. Concurrent work applies the same split-half logic as a performance ceiling in a different domain (crowd reading) [Floor-Ceiling]; our contribution is therefore the transplant of this reliability logic into RLVR selection evaluation and its pre-registered self-application, not the ceiling idea itself. To our knowledge, no prior work in this literature reports the reliability of its own oracle, and Section 4 shows that several of our initially promising comparisons no longer hold once it is reported.

3 Theory: one-sided corrections and what the monitored metrics cannot see

3.1 A 2×2 decomposition of stale-gradient estimators

For prompt x, response $Y = (A_1, \ldots, A_T)$ with terminal verifiable reward $R(Y)$, the current-policy gradient is $g_\pi(x) = \sum_t \mathbb{E}_{H_t, A_t \sim \pi}[Q_t^\pi(H_t, A_t)\, z_t]$ with $z_t = \nabla_\theta \log \pi(A_t \mid H_t)$. Throughout this subsection we assume a finite horizon T, absolute continuity ($\beta(Y) > 0$ whenever $\pi(Y) > 0$), exact unclipped likelihood ratios, and environment transitions that do not depend on the policy; implemented estimators that clip ratios for variance control are labelled as clipped variants wherever they appear. With behavior trajectories, token ratios $r_j = \pi(A_j|H_j)/\beta(A_j|H_j)$, prefix products $P_t = \prod_{j<t} r_j$ and suffix products $S_t = \prod_{j>t} r_j$, four population quantities arise from the same stale data:

$$
\begin{aligned}
g_{00} &= \textstyle\sum_t \mathbb{E}_\beta[r_t R z_t] && \text{(occupancy } \beta\text{, continuation } \beta\text{)} \\
g_{10} &= \textstyle\sum_t \mathbb{E}_\beta[P_t r_t R z_t] && \text{(occupancy } \pi\text{, continuation } \beta\text{)} \\
g_{01} &= \textstyle\sum_t \mathbb{E}_\beta[r_t S_t R z_t] && \text{(occupancy } \beta\text{, continuation } \pi\text{)} \\
g_{11} &= \textstyle\sum_t \mathbb{E}_\beta[P_t r_t S_t R z_t] = g_\pi.
\end{aligned}
$$

The subscripts state which axis is restored. The exact error decompositions

$$
\begin{aligned}
g_{10} - g_\pi &= \sum_t \mathbb{E}_{d_\pi, \pi}\big[(Q_t^\beta - Q_t^\pi) z_t\big] && \text{(continuation bias)}, \\
g_{01} - g_\pi &= \sum_t \big(\mathbb{E}_{d_\beta, \pi} - \mathbb{E}_{d_\pi, \pi}\big)\big[Q_t^\pi z_t\big] && \text{(occupancy bias)}
\end{aligned}
$$

show that the two one-sided estimators remove *different* biases; they are not substitutes.

For a unit direction u, write $s_c(x) = u^\top g_c(x)$, $c \in \{10, 01\}$, and let $\mathcal{I}_c^{(N)}$ denote everything retained by that one-sided score from N stale trajectories: its nonzero directional summands, their weights, and arbitrary summaries such as KL, ESS, norms, and self-similarity. A c-only selector is any possibly randomized top-k rule measurable with respect to $\mathcal{I}_c^{(N)}$; it does not use the omitted ratio axis or fresh target rollouts.

3.2 Sign reversal at arbitrarily small divergence

Theorem 1 (One-sided no-free-lunch)**.** *Fix $c \in \{10, 01\}$, $N \geq 1$, $0 < k < n$, and $\delta > 0$. There exist two binary-reward pools W_0, W_1 with different true top-k sets $S_0^\star, S_1^\star$ but*

$$\mathcal{L}_{W_0}(\mathcal{I}_c^{(N)}) = \mathcal{L}_{W_1}(\mathcal{I}_c^{(N)}), \qquad \max_{j\in\{0,1\}} \max_{x\in W_j} D_{\mathrm{KL}}(\pi_{j,x} \| \beta_x) < \delta.$$

Hence the c-only minimax risk obeys

$$\inf_{\widehat{S}:\,\mathcal{I}_c^{(N)}\text{-meas.}} \max_{j\in\{0,1\}} P_{W_j}(\widehat{S} \neq S_j^\star) \geq \frac{1}{2}.$$

The witnesses have unit normalized ESS and stale self-cosine, can make the selected directional gradient reverse sign, and remain valid after group normalization for every $K \geq 2$.

The canonical two-token witnesses make the reversal explicit: for $0 < \epsilon < 1/2$, each one-sided cell c has a construction with

$$u^\top g_\pi = -\frac{\epsilon}{2}, \qquad u^\top g_c = +\frac{\epsilon}{2}, \qquad D_{\mathrm{KL}}(\pi\|\beta) = 2\epsilon \log \frac{1/2+\epsilon}{1/2-\epsilon} = 8\epsilon^2 + O(\epsilon^4).$$

Appendix A gives both pools, the general top-k embedding, and the group-normalization proof.

Corollary 1 (Complete blindness of one-sided diagnostics)**.** *No $\mathcal{I}_c^{(N)}$-measurable diagnostic can uniformly certify the resulting top-k set. Certification requires information from the omitted axis, a structural assumption linking the axes, or fresh target outcomes.*

Proposition 1 (Cell disagreement in a witnessing construction)**.** *In the continuation counterexample,* $\cos(g_{10}, g_{01}) = -1$*: the two one-sided cells disagree maximally. Here g_{10} is reversed while g_{01} is exact; this is distinct from a* double *reversal (both one-sided estimators wrong against g_π), the event probed by the random search below.*

Empirically—an observation, not a theorem—in a seeded random search over 5×10^4 two-token, two-parameter environments we found *zero* instances of a double reversal (both one-sided estimators reversed against g_π) accompanied by cell agreement ($\cos(g_{10}, g_{01}) > 0.9$); the identity $g_{11} \equiv g_\pi$ was verified on every sample. This is a statement about the sampled proposal distribution, not a nonexistence claim: zero hits in 5×10^4 draws bounds the event rate under that distribution below 6×10^{-5} (rule of three, 95%), and environments outside the sampled family are not covered.

Absence of disagreement is not a safety guarantee: Theorem 1 already rules out every guarantee based on one one-sided information set, and general parameter sharing permits both bias vectors to have adverse projections without agreeing with each other. We therefore use cell disagreement only as an *unreliability signal*—a cheap reason to distrust a ranking—never as a selector or certificate. The random search is proposal-dependent empirical evidence for that warning, not a sufficiency theorem.

4 Measuring selection: floor, ceiling, and when comparison is meaningful

Estimand. All experiments target the decision practitioners consume: the top-k set by $s_i = \cos(\mu_i, v)$, where μ_i is the prompt's mean current-policy gradient and v a validation mean gradient. We report top-k precision against an on-policy oracle, boundary margins, and rollout budgets. Pointwise cosine averages are diagnostics, never primary results.

The split-half floor. The oracle itself is a K-rollout Monte Carlo estimate. We therefore report, for every run, the *corrected split-half floor*: rollouts are split into two disjoint halves (odd/even micro-groups), each half induces a top-k set (ties broken by *independent* per-half jitter; Section 7 explains why this correction matters), and the floor is the mean overlap fraction over 20 jitter pairs. Chance is k/n. Two scope notes. First, both halves score against the *same* validation gradient, so the floor is a rollout split-half agreement conditional on that validation vector, not an unconditional oracle reliability; shared validation error can inflate it. Second, the floor is not a lower bound on estimator performance — an estimator can measurably exceed it, so retention ratios (overlap divided by floor) can exceed 1 and are reported unclipped; their reading as "preserved fraction of available signal" holds only under the noise model of Proposition 2. Section 7 shows the floor is itself vulnerable to tie artifacts unless computed exactly this way.

A measurement ceiling. The floor does more than describe the oracle: it bounds what any comparison against that oracle can show.

Proposition 2 (Measurement ceiling)**.** *Let $t_i \overset{\text{iid}}{\sim} N(0, \tau^2)$, $a_i = t_i + e_{ai}$, $b_i = t_i + e_{bi}$, and $o_i = (a_i + b_i)/2$, with iid $e_{ai}, e_{bi} \sim N(0, \sigma^2)$ independent of t. For any nonlinear or randomized size-k rule $S(Z)$ satisfying $Z \perp (e_a, e_b) \mid t$,*

$$\mathbb{E}\frac{|S(Z) \cap \mathrm{TopK}(o)|}{k} \leq \mathrm{Ovl}_{n,k}\big(\sqrt{\rho_f}\big)\,,$$

$$\rho_h = \mathrm{corr}(a_i, b_i) = \frac{\tau^2}{\tau^2 + \sigma^2}, \qquad \rho_f = \mathrm{corr}(t_i, o_i)^2 = \frac{2\rho_h}{1 + \rho_h}.$$

Equality is attained by $S = \mathrm{TopK}(t)$. Since the population split-half agreement is $F = \mathrm{Ovl}_{n,k}(\rho_h)$, F determines the ceiling; finite-floor uncertainty is propagated through the same map.

Table 1 gives the mapping at our experimental geometry (n=512, k=51), validated by simulation (Appendix B). Two consequences organize everything that follows. First, at $\rho_h = 0$ the ceiling equals chance k/n exactly: every oracle-noise-oblivious selector has chance expected precision, so oracle-overlap evaluation contains no ranking information. By continuity, a near-chance population floor permits only a small excess over chance. Second, a selector at the ceiling has exhausted all expected overlap available to *any* method satisfying the same held-out noise condition; exceeding it requires either more reliable oracle rollouts or information statistically coupled to the held-out oracle noise.

Retention and its domain. When comparing regimes we report retention $(\text{prec} - c)/(\text{floor} - c)$ with c the chance level, declared *undefined* whenever $\text{floor} - c \leq 0.02$; saturated cells otherwise produce sign-flipping or exploding ratios (the 14B GSM8K cell below is the cautionary case).

Table 1: Population floor → model-ceiling mapping under Proposition 2 (n=512, k=51, chance 0.100; simulation, 600–800 replicates/cell). Plugging the v2 weak-signal floor estimates 0.137–0.176 into the model gives ceilings ≈0.30–0.37; finite-floor uncertainty must be carried through the same map.

population floor F	0.104	0.113	0.130	0.139	0.170	0.262	0.383
half-reliability ρ_h	0.02	0.05	0.10	0.12	0.20	0.40	0.60
ceiling on measurable precision	0.170	0.219	0.278	0.300	0.372	0.516	0.640

5 The regime map: where the flip does and does not happen

Setup. Qwen2.5-7B/14B-Instruct; GSM8K, MATH-500, and DAPO-Math-17k; drift between β and π is controlled by LoRA RFT for 50–400 steps; gradients are JL-projected; oracle uses K=32 fresh rollouts per prompt (8 micro-groups × 4); pools are n=256 (v1) or n=512 (v2) with k=10%. The v2 protocol was pre-registered after a full audit (equal per-cell budgets, seeded tie-breaks, independent jitter, manifests); judgment criteria were fixed before results were read. Two v2 seeds per task have completed at the time of writing; the decision rule is frozen, and the verdict will be recomputed under it when the remaining seeds arrive.

5.1 Where signal exists, one-sided selection can lose it

The v1 gate (7B GSM8K) passed its pre-registered criteria: one-sided estimators lose top-10% precision against the uncorrected baseline. At the largest drift (400 LoRA steps), the prefix-corrected estimator falls to precision 0.040 (overlap 1/25). The exact hypergeometric tail for this cell is $P(X \leq 1) = 0.27$, so we report it as a directional observation, not as an individually significant below-chance claim. One transparency note applies. The v1 floors were originally computed with the shared-jitter estimator that Section 7 later diagnosed as pitfall P2, which had inflated them to 0.32. The corrected re-estimation (Table 5) puts the drift-400 floor at 0.206. This is still a 2×-chance signal regime, and the 0.040 collapse lies far below both floor and chance; in other words, even the regime most favorable to our thesis had to survive our own guards. Downstream, fine-tuned accuracy after selection (base 0.740) spans 0.820–0.880 across oracle-, one-sided-, and random-selected pools, with no consistent ordering. In this saturation-adjacent regime, misranking has little room to change downstream accuracy, which is consistent with the results below.

On 7B MATH-500 the ordering flips. This is the strongest and least noisy signal regime we measured: the corrected floor is 0.288, and its K'-curve *rises* (Table 5). Here the uncorrected g_{00} is worst (0.160), and every stale estimator retains only 0.28–0.64 of the available signal. At 14B MATH-500 the loss concentrates on the suffix axis (signal retention g_{01}=0.44 vs. g_{10}=1.00), although the declining K'-curve there makes the absolute floor less certain. Which cell fails is therefore regime-dependent. What does not change is this: *some cell fails while the listed stale-only dashboard remains benign*—exactly the information restriction covered by Theorem 1, and no more.

5.2 Where signal is weak, the oracle limits resolvable gains

The corrected v2 runs put every completed condition at floors 0.137–0.176 against chance 0.100 (Table 2). Here stale overlap is above the random-selection null, although that test does not establish equivalence to the latent ranking. On GSM8K, the uncorrected stale baseline reaches precision 0.294/0.235 (seeds 1/2), with upper-tail exact $p = 2.8 \times 10^{-5}$ and 2.1×10^{-3}, and Fisher-combined $p = 1.0 \times 10^{-6}$. (The two seeds share the prompt pool; conditional on the pool, the hypergeometric

Table 2: v2 corrected results (n=512, k=51, chance 0.100; 2 seeds). Floors are corrected split-half (independent jitter). DAPO precisions are jitter-sensitive (±0.08) due to margin-0 tie degeneracy and shown as bands.

run	floor	g_{00}	g_{10}	g_{01}	g_{11}	regime
GSM8K s1	0.137	0.294	0.275	0.196	0.196	weak signal; stale at ceiling
GSM8K s2	0.176	0.235	0.196	0.216	0.216	weak signal
DAPO s1	0.176	0.04–0.16 (tie-degenerate)				margin 0.0; chance
DAPO s2	0.176	0.04–0.16 (tie-degenerate)				margin 0.0; chance

Table 3: Prompt-level sign reversal against the fresh oracle (completed v2 GSM8K seeds; zero-signal prompts excluded, denominators shown). Raw rates must be read against the reference row—the oracle's split-half self-disagreement on the same prompts—not against zero. Decision band = oracle ranks $k \pm 26$.

	overall reversal		decision-band reversal	
	s1	s2	s1	s2
g_{00}	67/163	80/168	12/37	12/34
g_{10}	68/163	73/168	13/37	10/34
g_{01}	61/163	63/168	10/37	8/34
g_{11} (full)	59/163	64/168	9/37	8/34
oracle self-disagreement (reference)	66/168	77/180	10/37	7/35

nulls are driven by independent rollout randomness, which is what the combination assumes.) The value 0.294 is close to the plug-in model ceiling (≈0.30 at floor 0.137; Table 1); a saturation claim requires propagating the finite-floor interval. At the point estimate, the stale estimator has exhausted nearly all overlap the model permits, so neither the losses of its one-sided competitors nor the gains of any other method are measurable in this regime. The DAPO pool is tie-degenerate: boundary margins are exactly 0.0, and precision itself moves by ±0.08 under tie-break jitter, so we report bands. There, all stale estimators are at chance, and so is everything else.

[PENDING — v2 seed 3, running] Additional GSM8K/DAPO seeds are being harvested; Tables 2 and 5 will gain one row group each and all seed-pooled statistics will be recomputed under the pre-registered rules. Verdict logic is frozen; only numbers change. *Reserved space: half a page (updated tables + two sentences).*

Prompt-level reversal prevalence (preliminary). Top-k overlap is a coarse summary. A reviewer will ask a finer question: how often does the sign of an individual prompt actually flip? Table 3 answers this for the completed v2 GSM8K seeds. Stale scores disagree in sign with the fresh oracle on 36–48% of signal-bearing prompts overall, on 24–35% of prompts inside the decision band, and on 6–15 of the oracle's top-51 prompts. These raw rates cannot be read against zero, because the oracle also disagrees with itself: an independent half of the oracle flips sign against the other half on 39%/43% of the same prompts (decision band: 27%/20%), as the reference row reports. Overall reversal for every cell lies within a few points of this reference. We therefore do not read the overall rates as estimator failure; they are dominated by oracle noise. The information lies in two contrasts that place the same oracle noise on both sides.

The first contrast concerns the decision band. There, the one-sided cells lie above the reference while the full correction lies on it: pooled band reversal is 0.34 (g_{00}) and 0.32 (g_{10}), against 0.24 for g_{11} and 0.24 for the reference, with g_{01} indistinguishable from full (0.25). Prompt-paired McNemar tests against g_{11} (exact, over all signal-bearing prompts) reach significance for g_{00} in seed 2 (27 vs. 11 discordant prompts, $p = 0.014$); the remaining paired contrasts are not significant ($p \geq 0.13$). Pooling is over a shared prompt pool, so we present these numbers as replication across seeds, not as independent evidence. This prompt-level resolution also exceeds what top-k precision offers in the same regime (Table 2), where the ceiling hides any separation.

The second contrast tests the theory's warning sign at the prompt level. Conditioned on the two one-sided cells disagreeing in sign, the prefix-corrected cell g_{10} is the reversed one in 59%/62% of cases (seeds 1/2), against a 37% reversal rate under agreement (Fisher exact $p = 0.015/0.007$). The same pattern replicates in the strong-signal v1 condition (7B, MATH-500): 81% against 45%, $p = 0.012$. In the saturated 14B regime, where selection itself degenerates (§5.3), the warning is correspondingly uninformative ($p \geq 0.44$); this is expected, because no signal remains there to protect. These results support the disagreement signal of Proposition 1 in real LLM runs, with one estimand caveat stated explicitly. For scalar per-prompt scores, a *double* reversal appears as cell *agreement*; the 37–45% reversal rate under agreement is exactly the case this warning cannot detect. This is consistent with "agreement guarantees nothing," and it is distinct from the proposition's vector-cosine statement. The DAPO seeds contribute only 2–4 signal-bearing prompts each (tie degeneracy, §5) and are excluded.

5.3 Where capability saturates, selection degenerates

At 14B on GSM8K ($\sim$95% accuracy), the pool contains 132/256 prompts with zero reward variance. Every estimator's precision equals the chance level (0.087–0.111 vs. 0.098), and the oracle disagrees *with itself* below chance (floor $0.080 < 0.098$). Selection, ranking, and certification therefore degenerate together: no method and no certificate can have an effect here. The practical consequence is that selection pipelines must scale candidate difficulty with model capability; otherwise they spend, without noticing, on machinery that cannot change the outcome.

6 A gap lower bound for exact certification

Even where a point estimate selects well, one may ask for the decision to be *certified*: spend fresh rollouts near the ranking boundary until confidence intervals separate the top-k set. We implemented this (sequential boundary refinement with shared-validation error propagation; pseudocode in Appendix C)—not as a method contribution, but as a measuring instrument. The inverse-square barrier is not specific to that instrument.

Proposition 3 (Exact top-k gap lower bound)**.** *Suppose a fresh query of prompt i returns an independent $N(\mu_i, \sigma^2)$ score. Let $\Delta = \mu_{(k)} - \mu_{(k+1)} > 0$ be the boundary gap. Any adaptive algorithm that is δ-correct for every mean vector, $0 < \delta < 1/2$, must satisfy on this instance*

$$\mathbb{E}[N_{(k)} + N_{(k+1)}] \geq \frac{2\sigma^2}{\Delta^2}\,\mathrm{kl}(1-\delta, \delta),$$

where kl *is binary relative entropy.*

The bound already holds in the two-boundary-prompt Gaussian subclass. The change-of-measure proof is in Appendix C; adaptive allocation can remove off-boundary waste but not the $\Delta^{-2}\log(1/\delta)$ dependence.

Our instrument returns the corresponding empirical negative. Across every completed condition (both scales, three tasks, drift 50–400, selection fractions 5–25%), certification fails at budgets 1.02–1.04× the uncorrected cost. The reason is geometric: top-k boundary margins are 0.00–0.24°, while achievable confidence radii are 51–180°. The gap is not a constant factor. With radius $\alpha_v(m) = \arcsin(c/\sqrt{m})$ under the instrument's iid Gaussian approximation, reaching the observed margins requires 4.1×10^5 to 1.0×10^8 times more observations—5–8 orders of magnitude in budget. These numbers are instrument-specific plug-in factors; Proposition 3 establishes that their inverse-square dependence is algorithm-independent in the Gaussian subclass. The measured failure is robust to (ϵ, δ)-PAC slack and to deepened validation. The cause is that prompt values are intrinsically dense at the selection boundary: *exact certification fails precisely where selection matters—at every margin we observed.* Approximate-membership and indifference-zone formulations are weaker demands our measurements do not exclude. Fresh rollouts spent on exact certification yielded no improvement in any condition we measured; Section 9 argues they belong in floor measurement instead.

7 Four evaluation pitfalls our own pipeline detected

Every mechanism below produced, at some point in this project, an apparently valid positive result that survived a first review and failed under a targeted audit. We report them as findings in their own right. Each comes with a guard now built into our protocol, and we suspect instances of each exist in the current rollout-reuse literature.

P1: Oracle-sample reuse (leakage). Our v1 hybrid intervention (completing the "missing half" with fresh continuations) showed a perfect 21/21-condition recovery, mean +0.46/+0.49 precision—which would have been the central causal result of the paper. A code audit found the treated cell reused rollouts that also defined the oracle; the shared noise, not causation, produced the recovery. A pre-registered equal-budget decisive experiment with disjoint samples showed *no consistent recovery* (e.g. GSM8K s1: fully-fresh cell 0.38 *below* mixed cell 0.46). *Guard:* strict sample disjointness between treatment and oracle (odd/even micro-group separation), enforced structurally.

P2: Shared-jitter tie inflation. Our first floor implementation broke score ties with a jitter shared across the two halves. On a tie-degenerate pool (DAPO, margins 0.0) this inflated the floor from 0.176 to 0.804, which briefly supported an incorrect "inverse correlation" conclusion. The same estimator had also inflated the v1 floors (0.32 reported vs. 0.15–0.21 corrected), which means every absolute floor in this literature computed with tie-unaware agreement warrants re-examination. *Guard:* independent per-half jitter streams, 20 seeded pairs, plus reporting precision *bands* under jitter on tie-heavy pools.

P3: Cell-budget imbalance. Comparing estimator cells with unequal effective rollout counts favors the cell with the larger budget without any visible sign of it; our v1 hybrid grid had this flaw in addition to P1. *Guard:* equal-K redesign of all cell comparisons, verified from manifests.

P4: Saturation-shared overlap inflation. An above-chance floor can be an artifact of *shared liveness* rather than shared ranking: prompts that saturate in both halves (all-correct or all-wrong) drop to an exactly-tied score, and the surviving "live" sets overlap between halves, producing top-k agreement with zero true ranking signal. In a synthetic pool with *no* ranking signal, saturation sharing alone yields floor 0.126 at K'=8, decaying toward chance (0.110) at K'=32. The signature

Table 4: K-scaling floor curves (observed exact recomputation; prediction by Spearman–Brown + Gaussian overlap). Chance 0.100. Note the *decreasing* observed curves—the P4 artifact signature—and the near-zero half-reliabilities. Verdict (pre-registered): structural absence.

run	r_1	K'=8	16	24	32	pred. 128	pred. 256
GSM8K s1	0.045	0.263	0.208	0.201	0.182	0.288	0.391
GSM8K s2	−0.005	0.220	0.182	0.170	0.156	0.094	0.094
DAPO s1	0.009	0.112	0.123	0.139	0.146	0.147	0.182
DAPO s2	−0.038	0.119	0.148	0.158	0.158	0.094	0.094

of this artifact is a floor that *decreases* as the oracle budget grows. Our measured K-scaling curves (Section 8) show exactly this decreasing shape, which means even our reported weak-signal floors are best read as upper bounds. *Guard:* always report the floor as a function of K'; treat a non-increasing floor curve as artifact evidence, and report tie mass alongside.

8 FIRST: diagnosing signal absence from existing artifacts

The pitfalls motivate the final contribution: FIRST, a pre-registered, artifact-only procedure (no new rollouts) that decides whether a pool supports selection *before* any new experiment is funded. It has three stages, all computed from artifacts every RLVR selection run already produces.

Stage 1: conditional-subset precheck. Any proposed pool intervention definable as a filter on existing prompts (e.g. "keep only live prompts, $0 <$ pass-rate < 1") already exists as a subset of completed runs. Computing the corrected conditional floor on that subset (with bootstrap CIs) prechecks the intervention at zero GPU cost: if the conditional floor does not clear $2\times$ chance in a majority of eligible runs, building the new pool has no evidential basis. On our pools this returned NO-GO, and the intervention (a live-only GSM8K pool) was cancelled without spending a GPU-hour.

Stage 2: K-scaling floor curve. From stored micro-group gradients the floor is *exactly* recomputed at K'=8, 16, 24, 32, and extrapolated to K'=64–256 by Spearman–Brown on the half-reliability r_1 mapped through the Gaussian overlap function, with a calibration check at the largest observed K'. The pre-registered verdict rule: if a majority of eligible runs are predicted to reach $2\times$ chance by $K' \leq 256$, recommend oracle expansion; otherwise declare structural absence.

Table 4 shows the result: one GSM8K seed extrapolates to $2\times$ chance at K'=128 from r_1=0.045; the other is flat at chance; both DAPO runs never reach it. Majority not met: *structural absence*—meaning no recoverable ranking signal *within the evaluated pools and budget range.* This is an epistemic verdict about what these pools and budgets can support, not an ontological claim that prompt-value differences do not exist.

Running the same analysis over every artifact-complete run in the project (twelve conditions including the v1 gate family; Table 5) widens the evidence base without changing the pre-registered vote. Signal regimes are rare (2–3 of 12). The two credible ones are exactly the conditions where the regime map's one-sided losses occur (7B MATH-500 and maximum-drift GSM8K). And half of all observed curves *decline* in K'—the P4 artifact signature—so absolute floors elsewhere are best read as upper bounds.

Table 5: Extended K-scaling evidence over all twelve completed conditions (corrected floors; "shape" is the observed $K'=8 \rightarrow 32$ direction, with $\downarrow$ marking the P4 artifact signature). The pre-registered verdict is computed only on the v2 rows and is unchanged; the extension shows how rare signal regimes are: three conditions clear 2× chance at $K'=32$, and only two of those with a rising curve. Floors here use random half-assignment resampling and may differ slightly from the fixed odd/even-split floors of Table 2.

condition	task	r_1	floor @ $K'=32$	reaches 2× chance	shape
v1 drift-50	GSM8K	0.020	0.177	pred. $K'=256$	$\uparrow$
v1 drift-100	GSM8K	0.053	0.146	pred. $K'=128$	$\downarrow$
v1 drift-200	GSM8K	0.067	0.147	pred. $K'=64$	$\downarrow$
v1 drift-400	GSM8K	0.065	**0.206**	at $K'=32$	$\uparrow$
v1 14B	GSM8K	0.018	0.113	never	$\downarrow$
v1 7B	MATH-500	0.070	**0.288**	at $K'=32$	$\uparrow$
v1 14B	MATH-500	0.075	0.221	at $K'=32$	$\downarrow$
v2 s1/s2	GSM8K	0.045/−0.005	0.182/0.156	pred. 128/never	$\downarrow\downarrow$
v2 s1/s2	DAPO	0.009/−0.038	0.146/0.158	never	$\uparrow\uparrow$
v2 s2	MATH-500	0.001	0.181	never	$\downarrow$

Stage 3 (replay): the ceiling made visible in budget space. As a final check we replayed all fresh-budget allocation policies on the completed v2 runs against a sample-disjoint truth (odd micro-groups only): stale cells, a pass-rate heuristic, random, uniform and boundary audits at 1–25% budgets, a sequential disagreement-guided audit, full-fresh at up to 100% budget, and our own pre-registered floor-gated protocol. The result is the measurement ceiling made concrete. On weak-signal GSM8K, no fresh budget yields an improvement. Full-fresh at 100% budget scores *lower* (0.167) than the zero-cost stale baseline (0.235). Our own pre-registered floor-gated protocol also loses (0.172); by its pre-registered loss condition, it is withdrawn from prescriptive use and kept only in the diagnostic role this section describes. The best zero-cost policy is not a gradient method at all but the behavior pass-rate heuristic (0.255). On DAPO, every estimator is at chance (0.08–0.16, truth margins 0.0). Throughout these replays, the monitored metrics stay in their normal ranges, precisely as Corollary 1 warns. The full correction's effective sample size is 0.1% (clip fraction 0.94–1.00); this is the variance-side reason one-sided truncations exist in the first place, measured in the same runs that show what the truncation costs.

Two pre-registered caveats accompany the method. First, the calibration check shows that the Gaussian model under-predicts observed floors at $K'=32$ (e.g. 0.094 predicted vs. 0.156 observed). This is consistent with the P4 artifact inflating the *observed* values, and it makes the verdict conservative in the safe direction. Second, the single positive extrapolation rests on an r_1 small enough to be affected by the P4 artifact, so we treat it as motivation for an inexpensive re-check when the remaining seeds arrive, not as grounds for a GPU expansion.

8.1 From diagnosis to decision: the FIRST protocol

The verdict is not the end of the pipeline. It is the first branch of a decision. Each regime in the map comes with a selection policy, and our own replays already measure the cost of each one. The diagnosis therefore extends into a decision protocol. The data-valuation literature encountered the same problem in supervised learning: when Shapley-based selection proved no better than random, the response was a corrected selection method, not abandonment of selection [Rethinking-Shapley, NASH].

The FIRST protocol inputs: the behavior rollouts the run already has (optionally a 48-prompt fresh probe); fix the threshold $\tau = 2\times$chance *before* looking.

1. **Measure.** Compute the corrected split-half floor (independent jitter) and the K-scaling curve with its Spearman–Brown budget estimate K^* (§8); run the four pitfall guards (§7).
2. **Branch.**
   (a) *No reachable signal* (floor $\approx$ chance, no finite K^*): abstain—sample uniformly and spend nothing on scoring. Every estimator we measured is at chance here, and the oracle disagrees with itself (§5.3).
   (b) *Weak signal* (chance $<$ floor $< \tau$): rank by the zero-cost behavior pass rate. In our replays it matched or outperformed every gradient estimator (0.255 vs. 0.235 best stale), and the measurement ceiling quantifies the limited expected oracle-overlap headroom in this regime (§4).
   (c) *Measurable signal* (floor $\geq \tau$): stale scores are usable. Compute both one-sided cells anyway, and send the prompts where they disagree in sign to a fresh audit (or drop them). Under disagreement the reversal rate was 59–81%; under agreement, 37–45% (three runs, $p \leq 0.015$). Agreement guarantees nothing—this is the case the warning cannot detect, and it is stated as such.
3. **Report.** Attach the floor, its uncertainty interval, the induced model-ceiling interval, and the branch taken to any downstream selection claim. A result above that interval falsifies at least one premise: the parallel-noise model, held-out-noise independence, or the evaluation pipeline.

The verdict costs no new rollouts; branch (c)'s audit spends fresh budget only on the disagreement set, which in our runs was 14–25% of signal-bearing prompts.

9 Practical guidance, limitations, and outlook

For practitioners. (1) *Measure the floor first*—run FIRST and take the branch its protocol prescribes (§8.1). A 48-prompt fresh probe is enough to estimate the split-half floor. Below $2\times$ chance (fix the threshold *before* looking), do not spend on selection. In our replays the best weak-signal policy was the zero-cost pass-rate heuristic, random was close behind, and no fresh budget up to 100% produced an improvement. (2) *Compute the boundary lower bound before budgeting certification*; for every pool we measured, the 5–8 order-of-magnitude plug-in gap rules out the available budget, and Proposition 3 shows that adaptive allocation cannot remove its inverse-square dependence. (3) *Scale candidate difficulty with capability*, or selection degenerates regardless of estimator. (4) *Report floor and ceiling with any selection comparison*; a claimed improvement above the measurement ceiling of its own oracle indicates assumption violation, estimator–oracle dependence, or an evaluation artifact. (5) *Distrust one-sided rankings when the two one-sided cells disagree*; agreement guarantees nothing, but disagreement is a zero-cost warning—formalized as branch (c) of the protocol.

Limitations. Our empirical evidence covers Qwen2.5 (7B/14B), math tasks, and LoRA drift, with two completed v2 seeds; the remaining seeds and a small-model capability–difficulty sweep are running, and the pre-registered verdict logic will be re-applied when they arrive. The ceiling proposition assumes iid latent Gaussian scores, equal-variance parallel oracle noise, and conditional independence from held-out oracle noise; simulation validates the resulting map at our geometry, and a distribution-free ceiling remains open. The disagreement result is an existence-plus-search finding, not a general sufficiency theorem. The theory counterexamples are two-token constructions by design: they show what the monitored metrics *cannot* exclude, not what typically happens, and the regime map is their empirical complement.

Safety relevance. Data selection is a curation channel: whatever a misranked estimator systematically promotes is amplified by subsequent training. Corollary 1 therefore has a safety reading. If stale selection promotes prompts on which shortcut or reward-hacking behavior scores well, the drift metrics a team monitors cannot flag the promotion, for the same reason they cannot flag the sign reversal itself. The selection layer is then an unaudited input to the reward loop. Theorem 1 transfers to a safety target whenever the same trajectory, support, and verifiable-reward assumptions hold. A safety application must separately audit partial observability and adversarial or misspecified rewards before using the measurement protocol.

Outlook. The negative results are central rather than incidental. They state the conditions under which stale-rollout data selection can work—unsaturated, tie-free, signal-bearing pools, which our diagnosis identifies at no extra cost—and the conditions under which it cannot. The measurement-first protocol (floor, ceiling, preregistration, and the four guards) applies to any selection-method paper, including ours.

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A Counterexample constructions and machine verification

Continuation construction. This reverses g_{00} and g_{10} in a two-token MDP. The first action $A_1 \sim \text{Bernoulli}(q)$, $q{=}1/2$, identical under both policies; reward $R = A_2$. Behavior success profile $Q^\beta(A_1{=}0) = \frac{1}{2} - \epsilon$, $Q^\beta(A_1{=}1) = \frac{1}{2} + \epsilon$; current policy swaps the two values. The gradient w.r.t. the logit of q is $g_\pi = -\epsilon/2$ while $g_{00} = g_{10} = +\epsilon/2$: the prefix already matches perfectly, so a perfect prefix ratio changes nothing, and the behavior-policy *ending* drives the sign. Mean suffix KL is $O(\epsilon^2)$; the 1-D cosine is -1 throughout.

Occupancy counterexample (reverses g_{00}, g_{01}). Behavior visits $P(A_1{=}0) = \frac{1}{2} + \epsilon$, current $\frac{1}{2} - \epsilon$; the second action is Bernoulli(1/2) under both, and $R = A_2$ if $A_1{=}0$, else $1 - A_2$. For the second-action logit, $g_\pi = -\epsilon/2$ while $g_{00} = g_{01} = +\epsilon/2$: no tokens remain after A_2, so the future ratio is exactly 1, and the stale *visit* distribution drives the sign. Trajectory KL is again $O(\epsilon^2)$.

In both cases the changed Bernoulli has

$$D_{\text{KL}}(\pi \| \beta) = 2\epsilon \log \frac{1/2 + \epsilon}{1/2 - \epsilon},$$

which is finite exactly for $0 < \epsilon < 1/2$ and has expansion $8\epsilon^2 + O(\epsilon^4)$.

Indistinguishable pools. We prove the $c = 10$ case; $c = 01$ is its time reversal. Add an exogenous initial branch $B \in \{\text{visible}, \text{hidden}\}$ with hidden probability h. In both branches $A_1 \sim \text{Bernoulli}(1/2)$ and the direction u controls only its logit. On the visible branch of prompt i, behavior and target share

$$P(A_2 = 1 \mid A_1 = 1) = \tfrac{1}{2} + \nu_i, \qquad P(A_2 = 1 \mid A_1 = 0) = \tfrac{1}{2} - \nu_i.$$

On the hidden branch behavior uses $P_\beta(A_2 = 1 \mid A_1) = 1/2$, while target world sign $s_i \in \{-1, +1\}$ uses the same display with ν_i replaced by $s_i\epsilon$. Reward is $R = A_2$. The retained prefix/current weight for the directional g_{10} summand is identically one, and

$$u^\top g_{10}(x_i) = \frac{(1-h)\nu_i}{2}, \qquad u^\top g_\pi(x_i) = \frac{(1-h)\nu_i + h s_i \epsilon}{2}.$$

Choose $0 < \nu_2 < \nu_1$ with $(1-h)\nu_1 < h\epsilon$. World W_0 takes $(s_1, s_2) = (+1, -1)$ and W_1 takes $(-1, +1)$. The strict one-sided ranking is $x_1 > x_2$ in both worlds, whereas the true top-1 swaps; in W_1 the selected prompt x_1 also has negative true directional gradient.

The behavior law, retained weight, and every retained directional summand are identical across worlds. The hidden target continuation has the same KL for signs $+1$ and -1 by Bernoulli symmetry, so supplying exact policy KL does not distinguish them either. The KL is $hO(\epsilon^2)$ and can be made below any δ. Therefore the full N-sample 10-only information has the same law for every N. If a randomized selector chooses x_1 with probability p, its errors in W_0, W_1 are $1 - p$ and p; their maximum is at least $1/2$. The same ambiguity applies to arbitrary top k by adding $k - 1$ prompts that dominate in both worlds and $n - k - 1$ prompts that are dominated in both.

For $c = 01$, let the direction control only the second-action logit, use $A_2 \sim \text{Bernoulli}(1/2)$ under both policies, and set $R = A_2$ after $A_1 = 0$ and $R = 1 - A_2$ after $A_1 = 1$. The visible branch shares $P(A_1 = 0) = 1/2 + \nu_i$; the hidden branch has behavior probability $1/2$ and target probability $1/2 + s_i\epsilon$. Then the retained current/suffix weight is one and the same two displayed gradients result. This proves the theorem for both axes.

Lemma 1 (Binary group normalization)**.** *Let $R \in \{0, 1\}$ satisfy $P(R = 1) = 1/2$, let z be a scalar directional score with $\mathbb{E}z = 0$, and define $g = \mathbb{E}[Rz]$. For $K \geq 2$ iid rollouts, standardize rewards by the within-group population standard deviation and assign update zero to homogeneous-reward groups. Then the expected normalized update is $C_K g$ for a constant $C_K > 0$ depending only on K.*

Proof. Let M be the number of successes and $\Delta = \mathbb{E}[z \mid R = 1] - \mathbb{E}[z \mid R = 0]$. Since $P(R = 1) = 1/2$ and $\mathbb{E}z = 0$, $g = \Delta/4$. Conditional on M, the expected group update is

$$\sqrt{(M/K)(1 - M/K)}\,\Delta.$$

Consequently

$$\mathbb{E}[g_{\text{group}}] = 4\,\mathbb{E}\Big[\sqrt{(M/K)(1 - M/K)}\Big]\, g = C_K g.$$

The expectation is strictly positive for every $K \geq 2$ because $P(0 < M < K) > 0$. Using sample rather than population standard deviation changes only the positive constant. All counterexample branches have success probability $1/2$, so signs and strict prompt order are preserved. □

B Measurement ceiling: derivation sketch and validation

Fix the latent vector t and define $p_i(t) = P(i \in \text{TopK}(o) \mid t)$. If $t_i \geq t_j$, exchangeability and equal noise variance imply $p_i(t) \geq p_j(t)$. One direct coupling swaps the iid noises of i and j: every realization in which only the lower-mean j enters the top k maps injectively to one in which only i enters. Therefore the size-k subset maximizing conditional expected overlap is $\mathrm{TopK}(t)$.

Given t, any $S(Z)$ satisfying $Z \perp (e_a, e_b) \mid t$ is merely a random mixture of size-k subsets. Hence

$$\mathbb{E}[|S(Z) \cap \mathrm{TopK}(o)| \mid t] \leq \sum_{i \in \mathrm{TopK}(t)} p_i(t).$$

Averaging proves the first inequality in Proposition 2. The parallel-measurement model gives $\rho_h = \tau^2/(\tau^2 + \sigma^2)$ and $\rho_f = \tau^2/(\tau^2 + \sigma^2/2) = 2\rho_h/(1 + \rho_h)$. Thus (t_i, o_i) is bivariate Gaussian with correlation $\sqrt{\rho_f}$, while (a_i, b_i) has correlation ρ_h, which gives the ceiling and population floor through the same Gaussian overlap function. This overlap is monotone in nonnegative correlation: for each top-k membership event A_i, its Gaussian noise stability has Hermite expansion $P(X \in A_i, Y \in A_i) = \sum_{m \geq 0} \rho^m \|f_{i,m}\|_2^2$, whose coefficients are nonnegative; summing over i gives $\mathrm{Ovl}_{n,k}(\rho)$.

Simulation at n=512, k=51 (600–800 replicates per cell; independent-jitter tie handling) produces Table 1. The table is a population-model map, not an equality for one finite observed overlap; in application we propagate a confidence interval for the floor through it. Heavy-tailed or heteroscedastic oracle noise can violate the model ceiling and must be tested separately. This distributional issue is distinct from the P4 saturation/tie artifact.

C Sequential boundary certification (measuring instrument)

Off-policy scores serve as a prior; every candidate receives a small fresh micro-group; scores carry confidence balls (shared validation error propagated); fresh rollouts are then spent only on candidates whose balls straddle the top-k boundary, until $\min L_{\text{selected}} > \max U_{\text{rest}}$ or the budget is exhausted, in which case *certification failure is reported, never hidden.* Variants (scalar CI, (ϵ, δ)-PAC with $\epsilon \leq 0.05$) behave identically in all our conditions because the binding constraint is the boundary margin itself (Section 6).

Proof of Proposition 3. Keep every non-boundary mean fixed and form an alternative instance by swapping $\mu_{(k)}$ and $\mu_{(k+1)}$. Let A be the event that the algorithm includes the original kth prompt. Uniform δ-correctness gives $P(A) \geq 1 - \delta$ on the original instance and $P'(A) \leq \delta$ on the alternative. Data processing from the adaptive transcript to the final decision gives

$$\mathrm{KL}(P_{\mathrm{tr}} \| P'_{\mathrm{tr}}) \geq \mathrm{kl}(1 - \delta, \delta).$$

Each observation from either boundary prompt changes its Gaussian mean by Δ and contributes $\Delta^2/(2\sigma^2)$ KL; observations from every other prompt contribute zero. The adaptive chain rule therefore gives

$$\mathrm{KL}(P_{\mathrm{tr}} \| P'_{\mathrm{tr}}) = \frac{\Delta^2}{2\sigma^2} \mathbb{E}[N_{(k)} + N_{(k+1)}],$$

which rearranges to the claimed lower bound.

D Preregistration and audit trail

All v2 judgment criteria, the precheck GO/NO-GO rule, the K-scaling verdict rule, and the reframing decision tree (“structural absence $\Rightarrow$ this paper’s structure”) were registered in the project log before the corresponding results were computed; the four pitfalls each carry a before/after commit pair. The verification script, manifests, and all readouts ship with the code release.